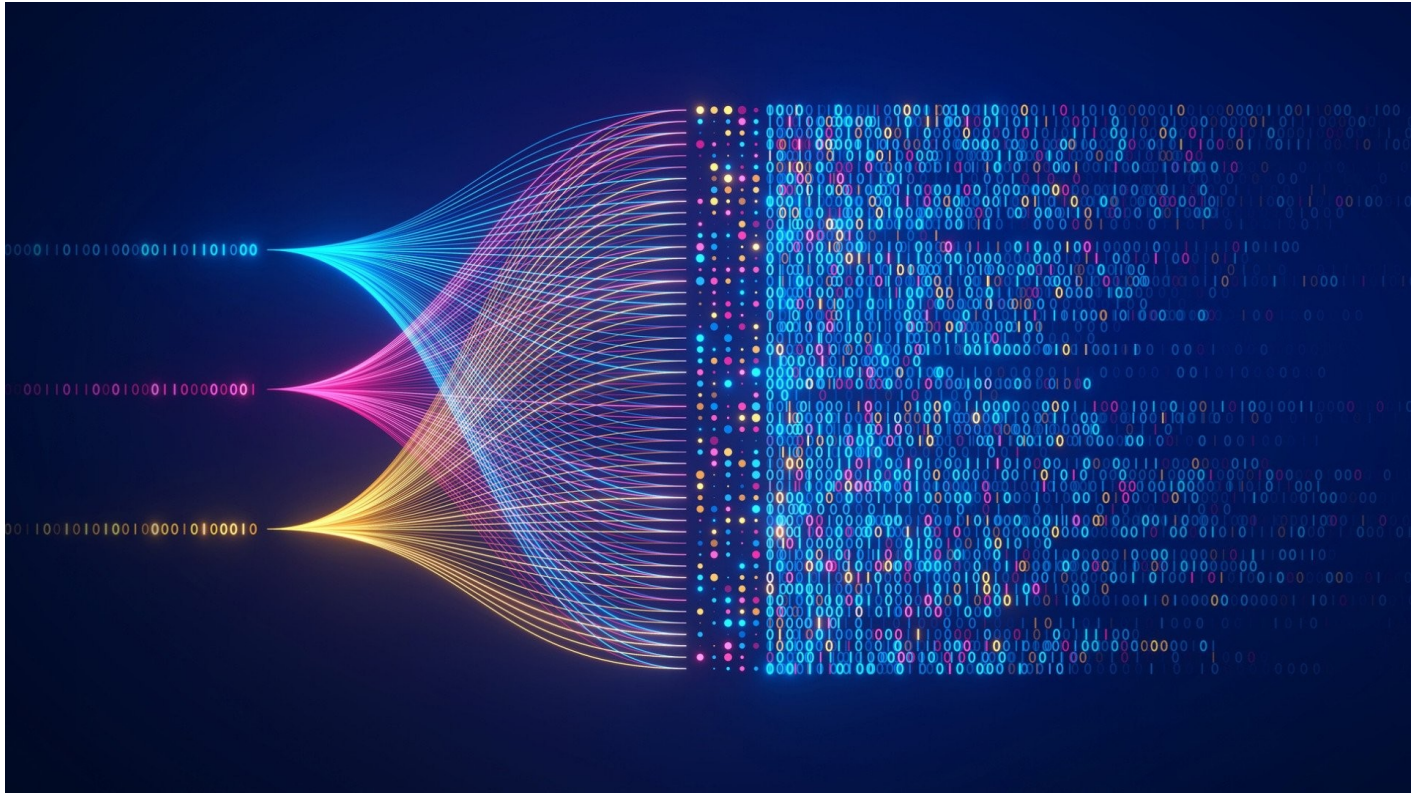




Local Large Language Model Lab

By Albert Zhang

Purpose:

This lab works through how to install and run a Local Artificial intelligence (AI) system, letting you have powerful generative AI capabilities that can be accessed offline and privately without concerns about privacy.

Background:

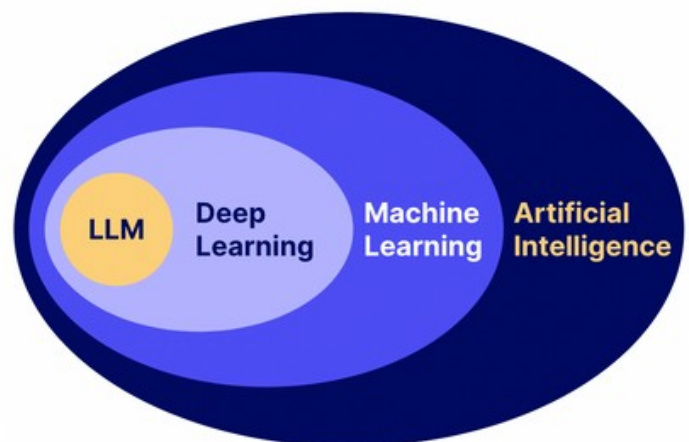
Introduction to AI

Artificial intelligence (AI) is a field in computer science that enables computers and machines to perform tasks that normally require human intelligence, the core engine powering this modern capability is the neural network. Neural networks can be considered the engine powering modern AI, inspired by the human brain, Neural networks are a type of model in machine learning, allowing computers to learn from complex patterns, simply put, neural networks are constructed with nodes connected by links, then organized in layers that act like processing units.

LLM

Definition:

LLM stands for Large Language Models, when people mention AI chat bots, they are all considered LLMs, for example, GPT-4, GPT-5 (by OpenAI), Gemini (by Google DeepMind), Claude (by Anthropic), LLaMA (by Meta), LLMs are a specific type of Generative AI. This is a highly specialized model trained only on massive amounts of text data, designed specifically for tasks like language understanding, conversation, and text generation.



Core Components (Transformer):

The core of an LLM is the Transformer, it is a type of neural network made to handle sequences of data. Transformers consist of layers that process input to understand relationships of words and the context all at the same time.

Tokenizer:

Before any text is processed by the LLM, it is broken into tokens; these tokens are smaller units like words, part of words, or just singular characters. Tokens are then converted into numbers for models to process, each token number is converted into a list of numbers (vectors), words with similar meanings end up with similar vectors, this lets the model to “understand” words better and form relationship between words through the vectors.

Self-Attention:

Self-attention is a crucial part of the Transformer. This mechanism allows the model to weigh the importance of each word in the input sequence, letting it understand the connections and relationships between words.

How LLMs work in practice:

Large language models are deployed in apps like: Chat bots, search assistants, code completion tools, content generation and data analysis and summarization.

Limitations and Risks of cloud LLMs:

LLMs still possess many limitations and risks such as:

Lack of long-term memory: LLMs treat each conversation like a separate task, meaning it cannot retain information across different chats.

Struggles with complex reasoning: LLMs have a hard time with complex reasoning tasks that require more than only understanding the literal meanings.

Static knowledge: since LLMs only have information up to their last training date, LLMs can't access real-time information.

Privacy Risks: Risks of sensitive information being used from input data from users

Local LLMs:**Why would you want to run your own local large language model?****Privacy and data security:**

Privacy and data protection is one of the biggest concerns of using the typical cloud-based LLM service, where the text you enter as a prompt and the response the model generates will most often be sent to internet servers owned by a third party. Depending on policies, this data might be stored and analyzed.

Offline Access and availability:

Cloud-based LLMs require stable internet connection to function, if your internet connection is sporadic, slow, or unavailable during traveling or power shortages, you will not be able to access cloud-based LLMs. In contrast to a Local LLM is entirely self-contained, meaning you can use the

anytime and anywhere, regardless of your internet connection. This provides a more reliable and accessible connection.

Customization:

When using a commercial cloud service, usually users are limited to specific models and configuration settings set by the provider of the cloud service. While running models locally lets you access much more of the model, where you are able to download any kind of open-source model that is fine tuned for specific tasks.

Ways to run a Local LLM:

Ollama - Ollama is one of the easiest ways to run local LLMs, it is a lightweight application + command-line tool that is specifically designed for downloading and running LLMs locally.

Docker - Docker is a platform that lets you package software into containers —lightweight, isolated environments that include everything the software needs to run container, like a mini-computer inside your computer.

Configuration:

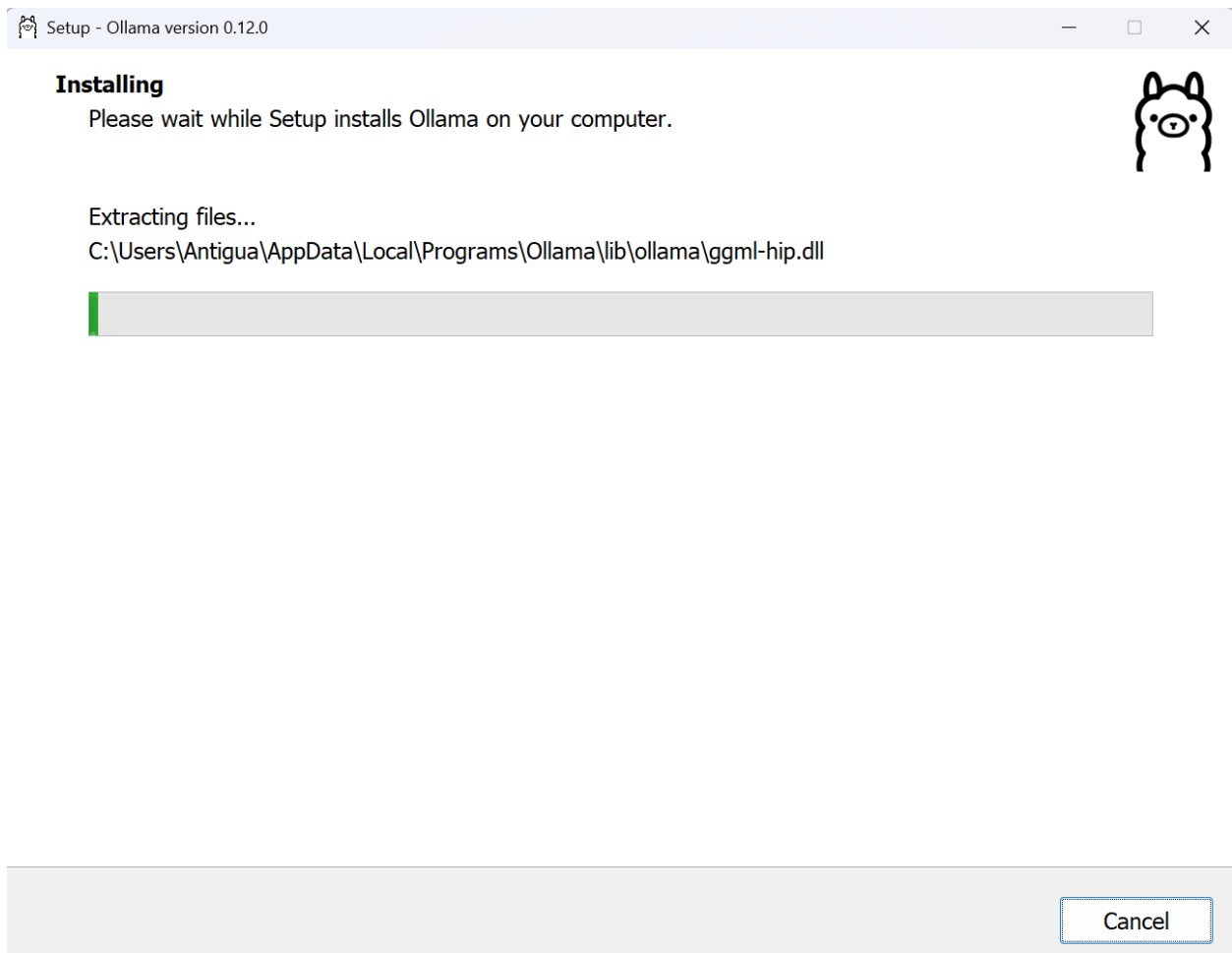
First, go to <https://ollama.com/>, select download, and then select the download for windows, finally, select “Download for Windows”



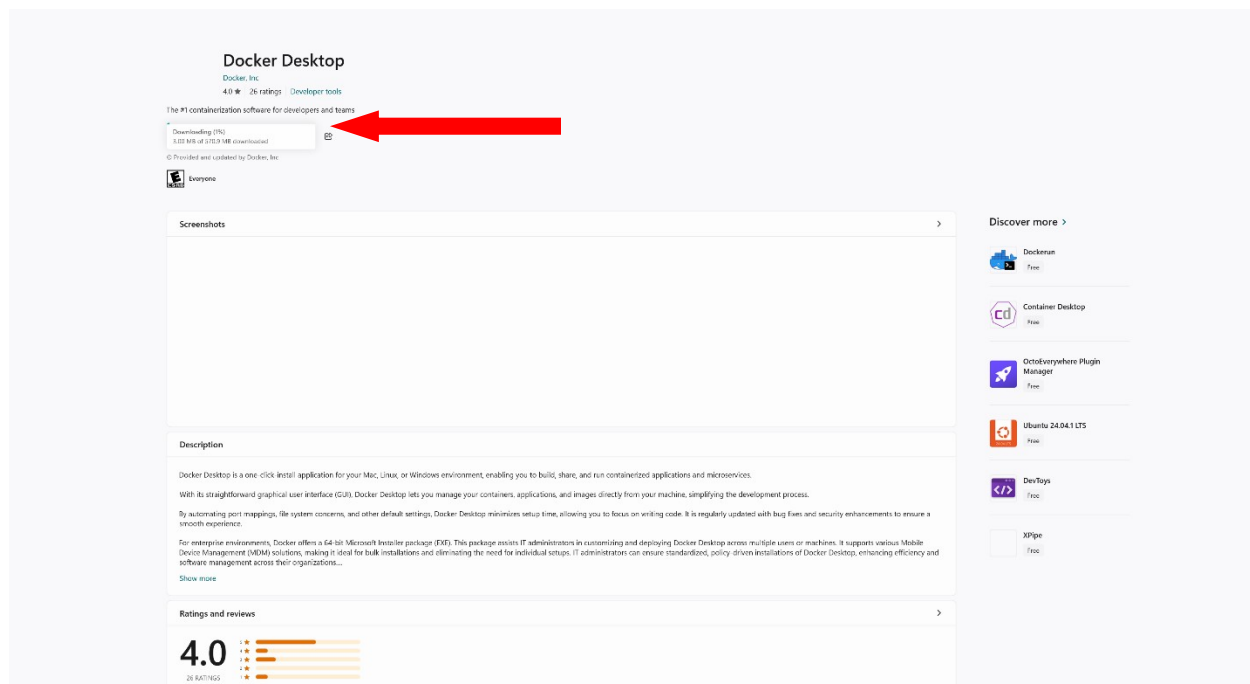
Download Ollama



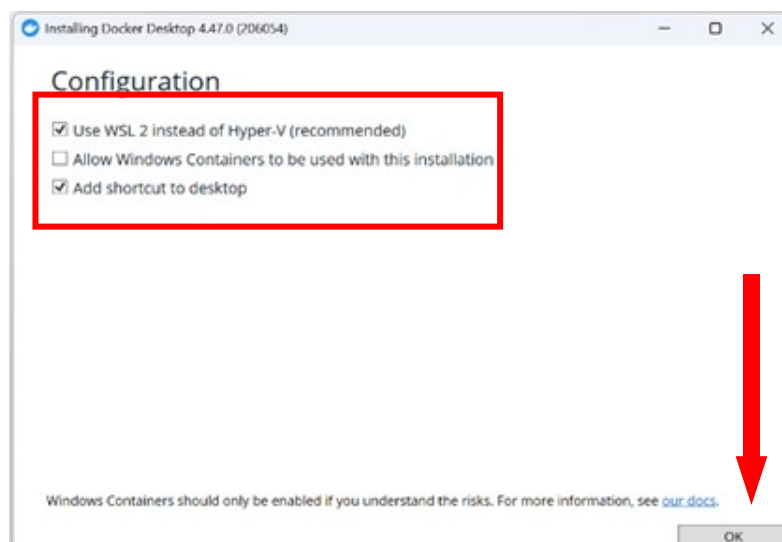
Run the downloaded file and Ollama will begin installing



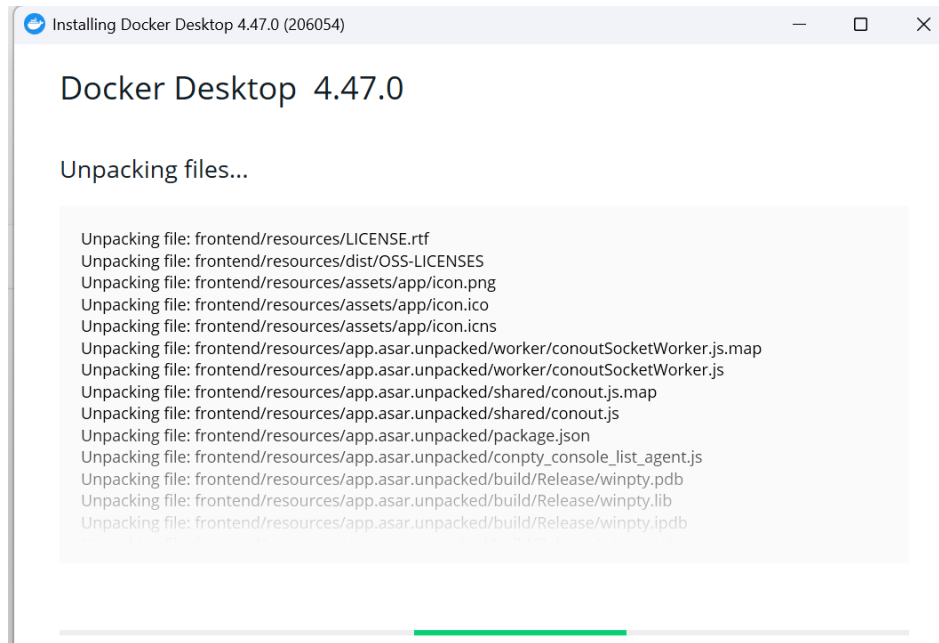
Search “Docker Desktop” in the Microsoft store and find the application by Docker. Inc, click download



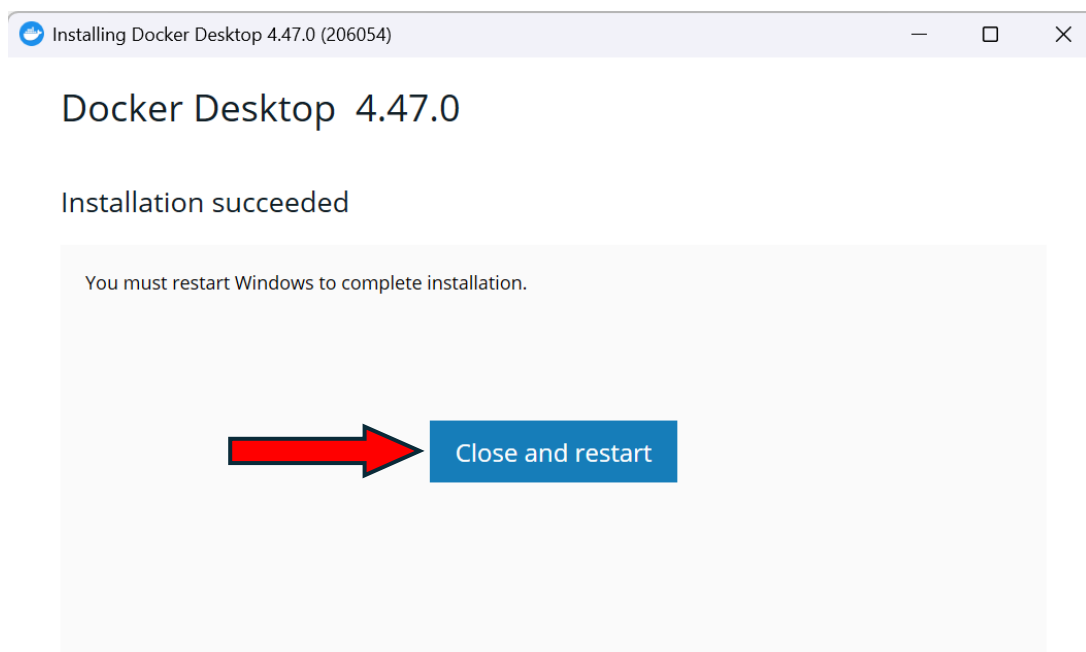
Afterwards you will be prompted with a configuration window, choose the first and third option:



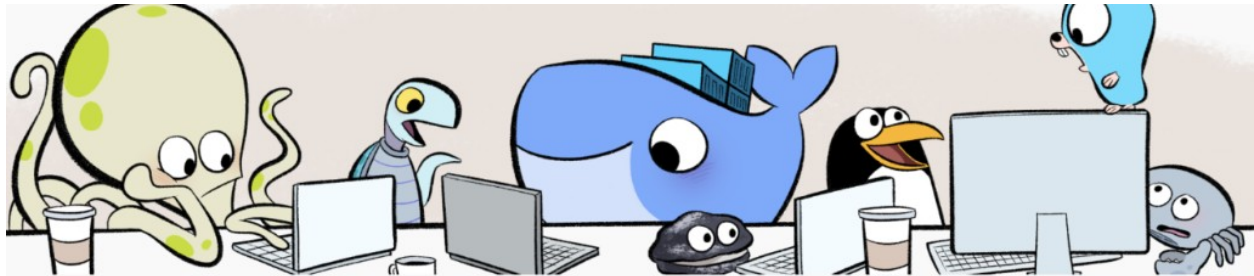
After Clicking “ok” this screen will appear



Select “Close and restart” after the unpacking files page.



After the restart, you will be prompted to accept the Subscription service agreement



Docker Subscription Service Agreement

By selecting **accept**, you agree to the [Subscription Service Agreement](#), the [Docker Data Processing Agreement](#), and the [Data Privacy Policy](#).

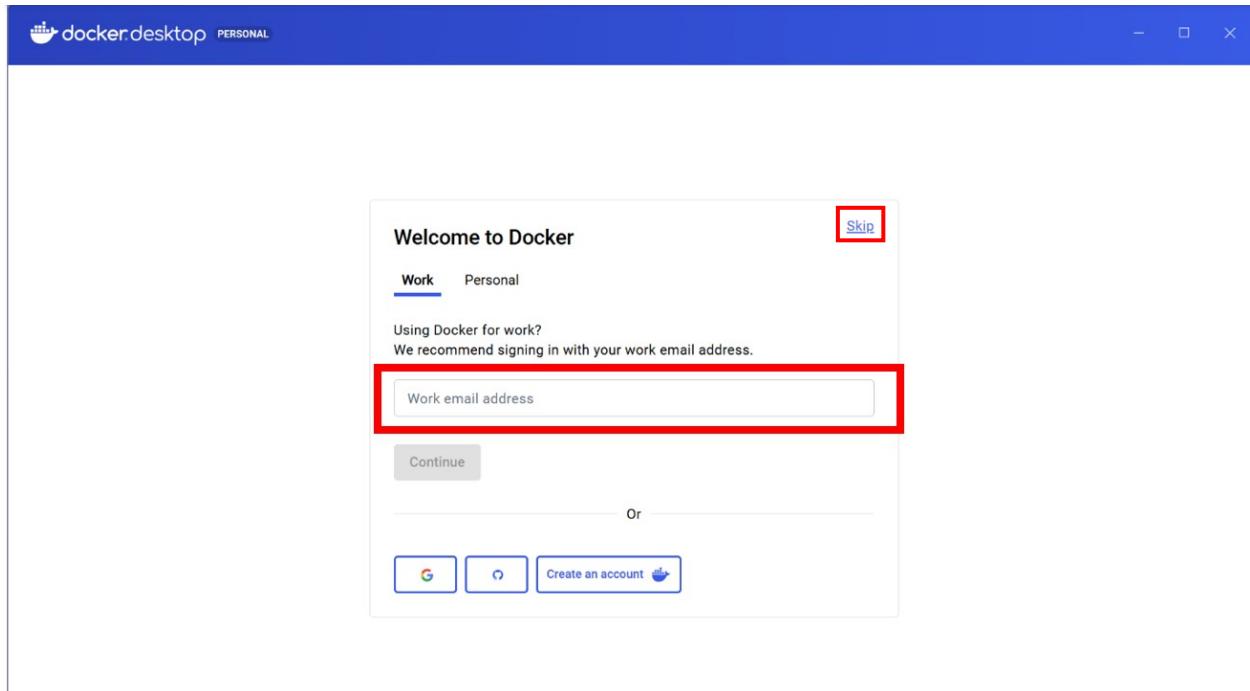
Note: Docker Desktop is free for small businesses (fewer than 250 employees AND less than \$10 million in annual revenue), personal use, education, and non-commercial open source projects. Otherwise, it requires a paid subscription for professional use. Paid subscriptions are also required for government entities. [Read the FAQ to learn more.](#)

[View Full Terms](#) 

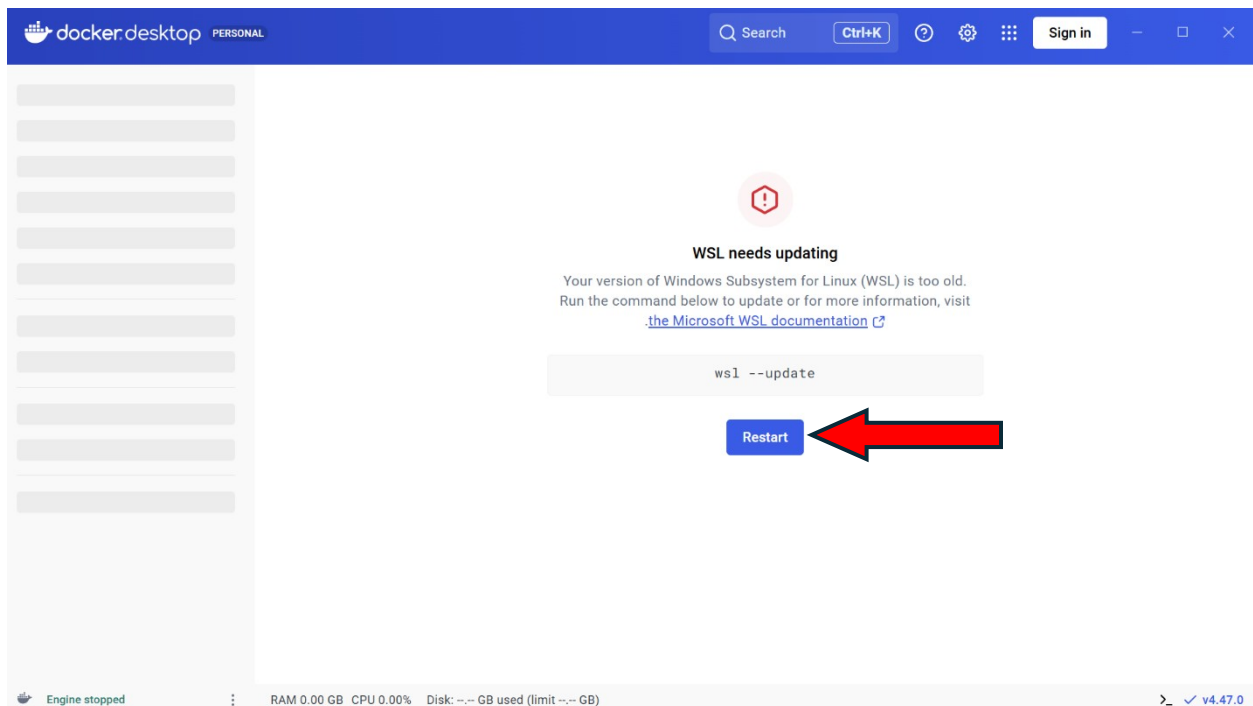
Accept

Close

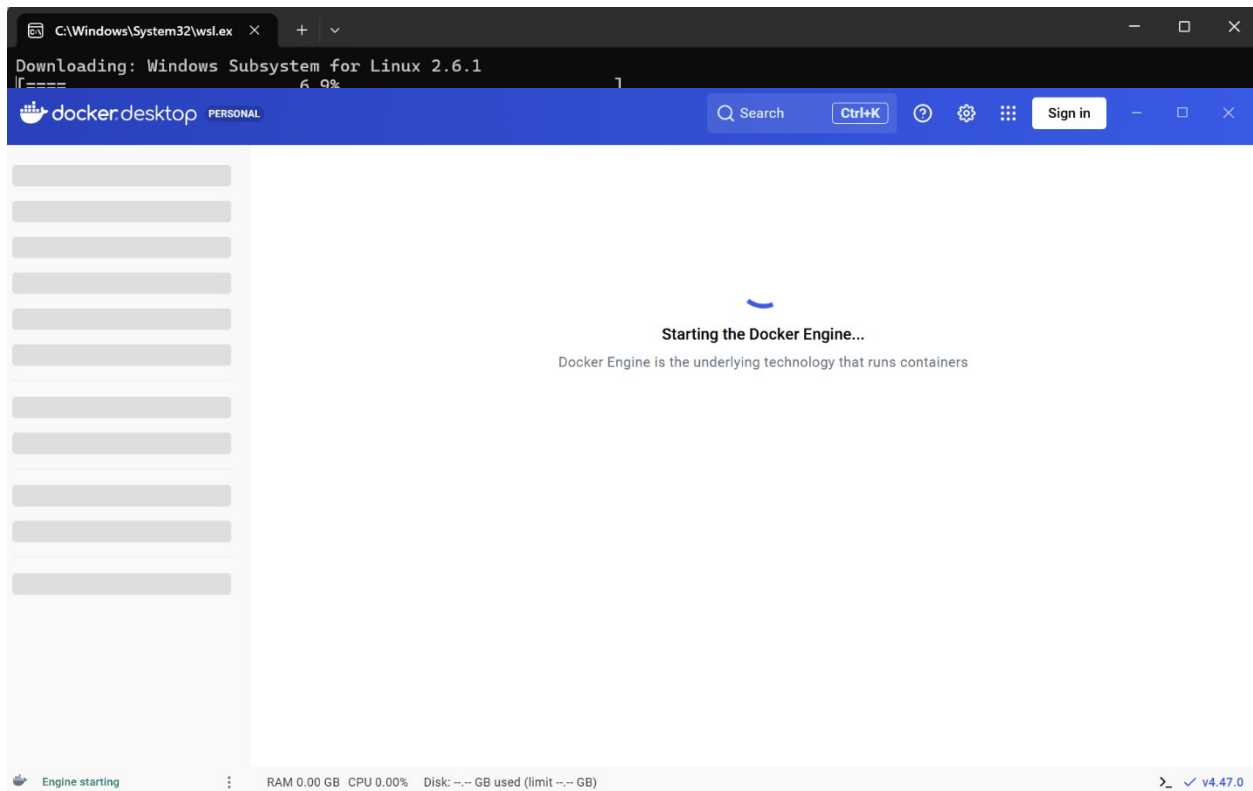
You will then be prompted to login or create an account, or you can choose to skip this step.



After an account is created, another restart will be required



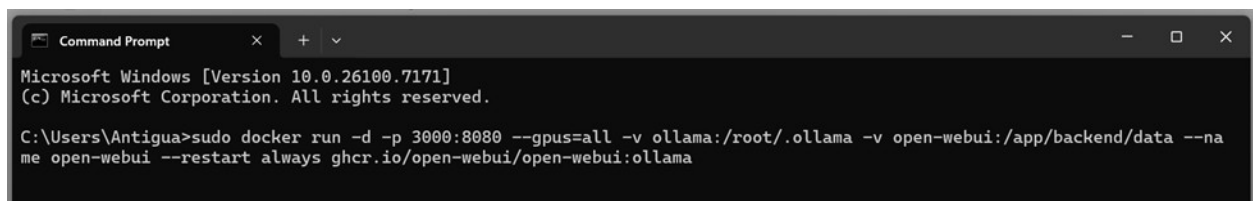
Window subsystem for windows will begin to install



After WSL is done installing, the docker engine will begin to start

Next, open the terminal in docker and enter the command:

```
Sudo docker run -d -p 3000:8080 --gpus=all -v ollama:/root/.ollama -v open-webui:/app/backend/data --name open-webui --restart always ghcr.io/open-webui/open-webui:ollama
```



```
Command Prompt
Microsoft Windows [Version 10.0.26100.7171]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Antigua>docker run -d -p 3000:8080 --gpus=all -v ollama:/root/.ollama -v open-webui:/app/backend/data --name op
en-webui --restart always ghcr.io/open-webui/open-webui:ollama
Unable to find image 'ghcr.io/open-webui/open-webui:ollama' locally
ollama: Pulling from open-webui/open-webui
1adabd6b0d6b: Pull complete
93300bbaa9a6: Pull complete
bb0d5f091034: Pull complete
2f382222568c: Pull complete
01f590d3d125: Pull complete
c7f3fe8e2df5: Pull complete
dc5c0390855f: Pull complete
a4df53ff8536: Pull complete
4f4fb700ef54: Pull complete
dbfb26191824: Pull complete
b9cbbb7560dc: Pull complete
36b48f89c8ab: Pull complete
c3750f149049: Pull complete
e1422ab5dc8c: Pull complete
114bd32fc7f4: Pull complete
54281cbcb0ed: Pull complete
Digest: sha256:d6e7e2dd06800229ddefdcfde55021d157a1689a6ee2ef16c42b083936ea388a
Status: Downloaded newer image for ghcr.io/open-webui/open-webui:ollama
a9a3f9dd4517719d01aa8e0e7ae4de60a604cd94064ec8733f02f8f980832623

C:\Users\Antigua>
```

After all the pulling is complete, you have successfully set up Web UI, to access it open to your web browser and navigate to `http://<your-host-IP>:3000`

To find your ip, open your command prompt by typing “cmd” in your search bar, and enter the “ipconfig” command-line, and you will find your ip.

```
Command Prompt
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :

Wireless LAN adapter Local Area Connection* 10:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . :
Link-local IPv6 Address . . . . . : fe80::ca12:b18b:11f
IPv4 Address. . . . . : 192.168.40.58
Subnet Mask . . . . . : 255.255.254.0
Default Gateway . . . . . : 192.168.40.1

Ethernet adapter Bluetooth Network Connection:

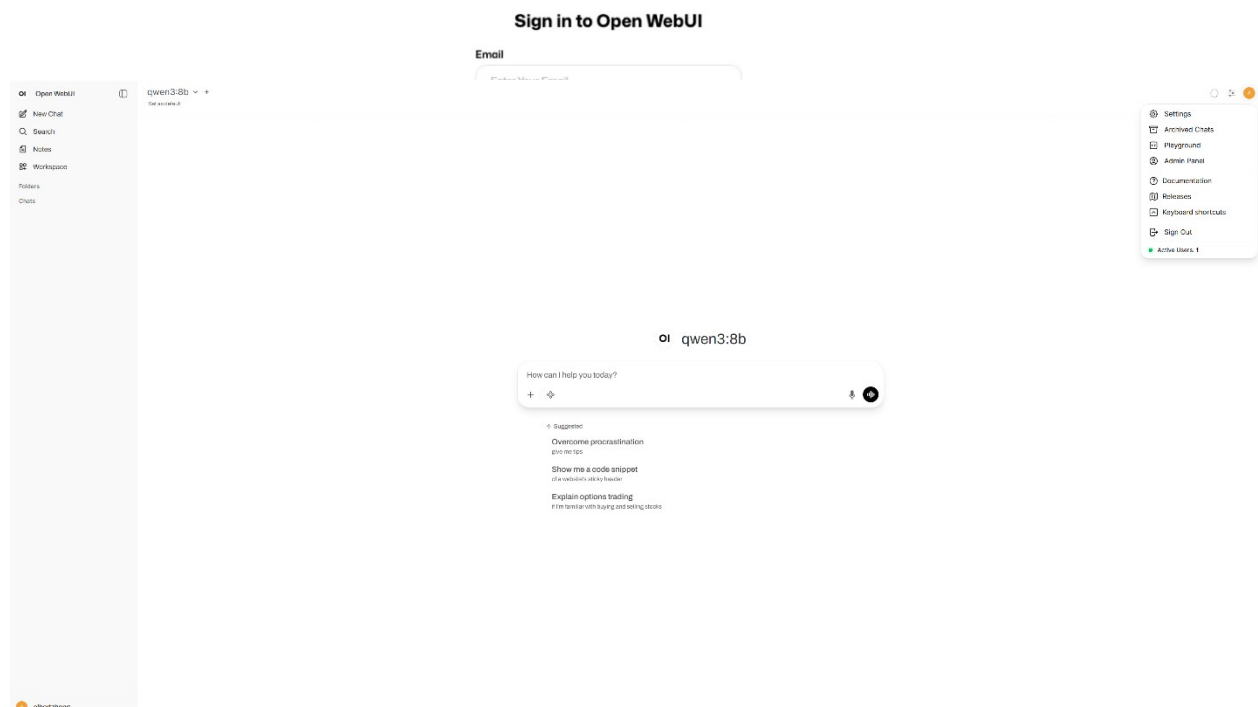
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :

Ethernet adapter vEthernet (WSL (Hyper-V firewall)):

Connection-specific DNS Suffix . :
Link-local IPv6 Address . . . . . : fe80::7a22:83ac:b58:5b42%48
IPv4 Address. . . . . : 172.24.80.1
Subnet Mask . . . . . : 255.255.240.0
Default Gateway . . . . . :

C:\Users\Antigua>
```

After navigating to the webpage, you will be prompted to sign in to your account and you will have successfully set up your WebUI for your local LLM



Problems:

Finding the most fitted model, VRAM or GPU memory is the primary bottleneck for running local large language models, most of the LLM's parameters must fit for smooth performance, 7B models need 4-8GB VRAM while a 13B model needs around 8-12GB VRAM.

An initial choice of use a large Multimodal LLM llava:34b, although this model was successfully loaded, some latency was occurring. The effective solution is to pivot to a smaller model like llava:7b, this allows the model to fully utilize the available VRAM, reducing latency. Despite llava:7b being a smaller model compared to more complex models, it remains highly effective and powerful, having all the core functions of any larger LLM.

Conclusion:

This lab demonstrates how a local large language model can set up locally, using tools such as Docker and Ollama. This is an uncomplicated and highly efficient method for running powerful models like Llama 3, gemma and other models directly on a local device. Running a large local language has many realistic advantages such as privacy that cloud based LLMs don't provide and being able to access powerful artificial intelligence, even while being offline.